

AARTHI JUJJAVARAPU

GenAI / AI Platform Engineer

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SUMMARY

AI/ML Engineer with 5+ years of experience building and deploying AI systems in financial services environments. Worked on GenAI copilots, RAG pipelines, document intelligence, and NLP applications using Python, Azure, and AWS. Hands-on experience with LLM workflows, inference optimization, model evaluation, and MLOps tooling, with a strong focus on reliability, monitoring, and production deployment.

PROFESSIONAL EXPERIENCE

Gen AI / ML Engineer | Citi Group - New York, USA (Contract)

Aug 2024 – May 2026

- Built a RAG-based internal document search platform using OpenAI embeddings, chunking strategies, and FAISS retrieval across a 500K-document corpus, reducing analyst research time by nearly 60% for compliance and operations teams.
- Fine-tuned LLaMA-2 and Mistral-7B models with QLoRA for financial summarization workflows, applying 4-bit quantization and ONNX optimization to reduce inference latency by 40% on A10G GPU deployments.
- Developed multi-step agentic workflows using LangChain and CrewAI for compliance document review, combining retrieval, entity validation, and structured JSON extraction to automate repetitive analyst tasks across 10+ workflow steps.
- Built a document intelligence service using LlamaIndex and Azure OpenAI for contract parsing and clause extraction, replacing a manual review workflow previously handled by a 10-member analyst team.
- Deployed FastAPI-based inference services on Azure AKS with Docker and autoscaling policies, implementing prompt caching and request batching to support 3x traffic spikes without impacting response latency.
- Created an internal LLM evaluation workflow to track faithfulness, relevance, and hallucination metrics, helping teams improve prompt quality and reduce unreliable responses during model iteration cycles.
- Worked on RLHF-based reward modeling using feedback collected from internal SMEs to align financial summarization outputs with compliance and governance expectations.
- Added responsible AI guardrails including PII filtering, response validation, and SHAP-based explainability checks to support internal risk management and AI governance requirements.

AI / ML Engineer | Fidelity Investments — Boston, USA (Contract)

Nov 2022 – May 2024

- Fine-tuned GPT and BERT-based NLP models for earnings call summarization, sentiment analysis, and entity extraction workflows, improving evaluation F1 scores by 11 points and helping research teams process financial reports faster.
- Built RAG-based internal assistants using Pinecone and Weaviate for enterprise knowledge retrieval workflows, reducing average query response latency from 4.2 seconds to 1.1 seconds through retrieval tuning, embedding optimization, and caching strategies.
- Maintained FastAPI inference services handling 50K+ daily predictions with Prometheus and Grafana monitoring for latency, drift, and reliability tracking, helping maintain 99.9% service uptime across production workloads.
- Deployed transformer-based document classification models on AWS SageMaker to automate support ticket routing across business teams, reducing request misrouting by 34% for more than 200K monthly support cases.
- Developed Spark and Databricks preprocessing pipelines for over 50M financial documents stored in BigQuery and Snowflake, adding automated validation checks to detect silent data drift before downstream model training.
- Applied LoRA and PEFT-based fine-tuning techniques for financial NLP workloads, reducing GPU training costs by nearly 60% while improving iteration speed for model updates.

- Optimized customer-facing NLP inference APIs using ONNX export and TensorRT compilation on GPU-backed deployments, reducing p99 latency from 380ms to 140ms and improving application responsiveness.
- Managed end-to-end MLOps workflows using MLflow, Airflow, and DVC for experiment tracking, retraining orchestration, and dataset versioning, enabling weekly production releases with minimal deployment downtime.

ML Engineer | Zensar Technologies — Pune, India

May 2021 – Jun 2022

- Built and deployed machine learning models using XGBoost and scikit-learn for customer churn prediction workflows, exposing predictions through Flask REST APIs integrated into production business applications.
 - Developed time-series forecasting pipelines using ARIMA, Prophet, and LSTM models for demand planning use cases, with scheduled retraining workflows through Airflow based on updated business data.
 - Designed Spark and Pandas-based ETL pipelines to process structured and unstructured datasets stored in BigQuery and Snowflake, handling batches exceeding 10M records for downstream analytics and model training.
 - Built a real-time anomaly detection workflow for transaction monitoring using isolation forests and LSTM autoencoders, reducing false-positive alert rates by 22% and improving investigation efficiency for operations teams.
 - Deployed containerized REST APIs on AWS EC2 using Docker and Kubernetes while implementing CI/CD pipelines through GitHub Actions to automate testing, deployment, and release workflows.
 - Worked with cross-functional business and analytics teams to translate operational requirements into ML solutions, presenting model insights and forecasting outputs to support data-driven decision making.
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KEY PROJECTS

TraceChain — AI Workflow Observability & Reliability Framework

- Built an SDK-first observability framework for AI workflows using decorator-based instrumentation to capture trace spans, retry diagnostics, token usage, and streaming events without requiring changes to existing application logic.
- Published Python and TypeScript SDKs to PyPI and npm with OpenTelemetry-based tracing support, helping developers integrate workflow observability into LLM and agentic AI applications with minimal setup effort.
- Developed a FastAPI backend with PostgreSQL storage and WebSocket event streaming to process workflow traces in real time and support debugging across multi-step AI pipelines.
- Built a Next.js dashboard with ReactFlow-based agent execution graphs, workflow timelines, and retry diagnostics, improving visibility into latency bottlenecks and tool-call failures during agent execution.
- Designed the platform with a self-hosted and local-first architecture using Docker-based deployment workflows, supporting teams working with sensitive or regulated AI data where external telemetry sharing was restricted.

Stack: Python, FastAPI, PostgreSQL, Next.js, OpenTelemetry, Docker, ReactFlow, WebSockets

AI Career Assistant- Portifoli/demo

- Built and deployed an AI-powered career assistant application on Hugging Face Spaces using RAG pipelines and LLM orchestration, enabling resume analysis, interview preparation, job search assistance, and interactive career-related query handling through a web-based interface.

Inference Cost Optimization

- Implemented GPU autoscaling and spot-instance based deployment strategies on AWS for ML inference workloads, reducing monthly infrastructure costs by nearly 35% while maintaining stable throughput and application performance during peak traffic periods.
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TECHNICAL SKILLS

Languages	Python, SQL, Bash, Go, Java
LLMs & GenAI	GPT-4, LLaMA-2/3, Mistral, Gemma · Fine-tuning: LoRA, QLoRA, RLHF, PEFT
Frameworks	PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, OpenAI API
Agentic / RAG	CrewAI, LangGraph, FAISS, Pinecone, Weaviate, Chroma
Inference & Optim.	ONNX, TensorRT, vLLM, 4-bit quantization, knowledge distillation
MLOps	MLflow, Airflow, Kubeflow, DVC, Weights & Biases, Prometheus, Grafana
Cloud & Infra	AWS (SageMaker, EKS), Azure (AKS, Azure OpenAI, Azure ML), GCP (Vertex AI)
Serving & APIs	FastAPI, Flask, Docker, Kubernetes
Data Engineering	Spark, Databricks, Delta Lake, Kafka, Pandas, BigQuery, Snowflake
DevOps / CI/CD	Git, GitHub Actions, Jenkins, Terraform
Eval & Safety	LLM eval frameworks, SHAP, LIME, PII filtering, guardrails
Observability	OpenTelemetry, Prometheus, Grafana, structured tracing

EDUCATION

M.S. in Computer Science · University of Texas at Arlington May 2024

CERTIFICATIONS

TensorFlow Developer Certificate · Microsoft Certified: Azure AI Fundamentals · IBM AI Engineering Specialization (Coursera) · Microsoft Certified: Data Analyst Associate